DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • PES UNIVERSITY, BENGALURU

UE24CS395: Database Management Systems Mini Project • Academic Year 2025–2026

Relational DML Operations & ACID Transactions

Evaluation Matrix Row 6: DML Operations • Weightage: 4 Marks

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1. Comprehensive DML Implementation

We implemented data manipulation statements in `database/sql/dml-examples.sql` covering record insertion, updates, cascade deletions, complex data retrieval, and multi-statement ACID transactions.

2. Multi-Table INSERT Operations

The listing below demonstrates multi-table record provisioning across core entities and associative junction bridges.

SQL INSERT Statements across Relations

```
-- Provisioning primary investigative case
INSERT INTO cases (case_id, title, description, case_type, status, priority, lead_officer_id, incident_date)
VALUES ('CASE-2026-088', 'Operation Iron Falcon', 'Illicit arms smuggling ring operating in outer ports',
        'Arms Trafficking', 'Open', 'High', 'OFF-001', '2026-10-15');

-- Associating implicated suspect via junction table case_people
INSERT INTO case_people (case_id, person_id, role_in_case, notes)
VALUES ('CASE-2026-088', 'PER-001', 'Key Financier', 'Identified at customs container checkpoint');

-- Seizing digital exhibit linked to case
INSERT INTO evidence (evidence_id, title, evidence_type, description, case_id, collected_by_officer_id, barcode)
VALUES ('EVD-088', 'Encrypted Satellite Transceiver', 'Digital Hardware', 'Found inside container crate', 'CASE-2026-088', 'OFF-001', 'BC-882910');
```

3. Multi-Row UPDATE Operations

SQL UPDATE Statements

```
-- Elevate investigation priority and update timestamp
UPDATE cases
SET priority = 'Critical', status = 'Under Investigation', updated_at = CURRENT_TIMESTAMP
WHERE case_id = 'CASE-2026-088';

-- Update storage location of seized exhibit upon transfer to vault
UPDATE evidence
SET storage_location = 'Central Forensic Vault 4B', status = 'Secured In Vault', updated_at = CURRENT_TIMESTAMP
WHERE evidence_id = 'EVD-088';
```

4. DELETE Operations and Foreign Key Cascades

Our schema uses `ON DELETE CASCADE` for associative bridges and `ON DELETE RESTRICT` for evidentiary custody records.

SQL DELETE Demonstration with Referential Verification

```
-- Verify junction record exists before deletion
SELECT COUNT(*) FROM case_people WHERE case_id = 'CASE-2026-088'; -- Returns 1

-- Delete parent case record
DELETE FROM cases WHERE case_id = 'CASE-2026-088';

-- Verify cascade deletion automatically purged child junction rows
SELECT COUNT(*) FROM case_people WHERE case_id = 'CASE-2026-088'; -- Returns 0
```

5. ACID Transaction Demonstration with SAVEPOINT

To demonstrate transaction management, we executed a multi-statement transaction incorporating `BEGIN`, `SAVEPOINT`, `ROLLBACK TO SAVEPOINT`, and `COMMIT`.

ACID Transaction Execution Script (database/sql/dml-examples.sql)

```
BEGIN TRANSACTION;

-- Step 1: Provision new test officer
INSERT INTO officers (officer_id, full_name, badge_number, contact_email, role)
VALUES ('OFF-TEMP-01', 'Cadet Vikram Rao', 'CID-TMP-01', 'vikram.rao@police.gov.in', 'Analyst');

-- Step 2: Establish savepoint prior to tentative case assignment
SAVEPOINT pre_case_provisioning;

-- Step 3: Insert tentative case record
INSERT INTO cases (case_id, title, case_type, status, priority, lead_officer_id)
VALUES ('CASE-TEMP-01', 'Tentative Lead Inquiry', 'Fraud', 'Open', 'Low', 'OFF-TEMP-01');

-- Step 4: Simulate operational abort on case creation (Rollback to savepoint)
ROLLBACK TO SAVEPOINT pre_case_provisioning;

-- Step 5: Finalize transaction
COMMIT;
```

Terminal Verification Output: Transaction State After COMMIT

```
SELECT officer_id, full_name, role FROM officers WHERE officer_id = 'OFF-TEMP-01';
 officer_id  |    full_name     |  role
-------------+------------------+---------
 OFF-TEMP-01 | Cadet Vikram Rao | Analyst
(1 row: Proves officer insert was preserved after COMMIT)

SELECT case_id, title FROM cases WHERE case_id = 'CASE-TEMP-01';
 case_id | title
---------+-------
(0 rows: Proves tentative case was cleanly rolled back to SAVEPOINT)
```